

Project Proposal

Software for Measuring Area of Irregular Shapes from Images

Delivery Dead line: Friday May 8th 2026

1. Project Title

Development of an Image-Based Irregular Shape Area Measurement Software

2. Project Description

Students (group of up to 4 students) are required to design and develop a **software application** (web, mobile, or desktop) capable of calculating the **area of irregular homogeneous shapes from images**, similar to the functionality provided by **ImageJ** software.

The software should allow users to:

- Upload an image
- Define scale (real-world calibration)
- Select or segment an irregular shape
- Automatically compute and display the area

This project combines **image processing, user interface design, and software engineering** to solve a real-world problem. Students may follow the demo video that achieve the same task using Image J software. Sample images are provided for test. Students should apply their solution to those images and verify the results using a macro written in Image J software as the provided video.

3. Objectives

By completing this project, students will:

- Understand basic **digital image processing concepts**
- Implement **image segmentation and edge detection techniques**

- Convert pixel measurements into **real-world units**
 - Design an intuitive **user interface**
 - Gain experience in **software development and deployment**
 - Work collaboratively in teams
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4. Functional Requirements

The developed software must include:

Core Features

- Image upload (JPG, PNG, etc.)
- Region selection (manual or automatic)
- Calibration tool (set scale using known reference length)
- Area calculation of selected region
- Display results in standard units (e.g., cm², mm²)

Optional Advanced Features (Bonus)

- Automatic object detection
 - Multiple region measurement
 - Export results (CSV / PDF)
 - Real-time camera input (mobile apps)
 - AI-based segmentation
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5. Technical Requirements

Students may choose one platform:

Option 1: Web Application

- HTML, CSS, JavaScript
- Frameworks: React, Vue, or Angular
- Libraries: OpenCV.js

Option 2: Mobile Application

- Android (Java/Kotlin)
- Flutter / React Native
- OpenCV pytorch mobile libraries

Option 3: Desktop Application

- Python (Tkinter / PyQt)
 - C# (.NET)
 - C++ with OpenCV
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6. Suggested Methodology

Step 1: Image Acquisition

- Load or capture image

Step 2: Preprocessing

- Grayscale conversion
- Noise reduction
- Thresholding

Step 3: Segmentation

- Edge detection (Canny)
- Contour detection
- Region selection

Step 4: Calibration

- User defines a known length in the image
- Convert pixels $\rightarrow$ real-world units

Step 5: Area Calculation

- Count pixels inside selected region
 - Convert to real area
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7. Project Deliverables

Each group must submit the following:

1. Software Application

- Fully functional program (web, mobile, or desktop)

2. Source Code

- Clean, well-documented code
- Organized project structure

3. Voice-Over Demonstration Video

The video must include:

- Project overview
- Explanation of how the software works
- Demonstration of features
- Example use case
- Comparing the results with that obtained from Image J software

4. Documentation

- System design
- Algorithms used
- Challenges faced
- Future improvements

5. Google Drive Submission

All materials must be uploaded to a shared Google Drive folder including:

- Source code
 - Executable / APK / hosted link
 - Video presentation
 - Documentation
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8. Team Structure

- Students will work in **groups (2–4 members)**
 - Each member must contribute to:
 - Development
 - Testing
 - Presentation
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9. Evaluation Criteria

Criteria	Weight
Functionality & Accuracy	30%
User Interface & Usability	15%
Code Quality & Documentation	20%
Innovation / Extra Features	10%
Video Presentation	15%
Completeness of Submission	10%

10. Timeline (Suggested)

Week	Task
1	Proposal & planning
2–3	Development
4	Testing & debugging
5	Final submission & video

11. Expected Learning Outcomes

At the end of this project, students will be able to:

- Apply **image processing techniques in real applications**
 - Translate theoretical concepts into working software
 - Collaborate effectively in teams
 - Present technical work professionally
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12. Notes

- The software should aim to **replicate the core functionality of ImageJ** but does not need to match it completely.
 - Creativity and additional features are highly encouraged.
 - Clear explanation in the video is essential for grading.
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